

NOTES

IDLE

Hello!!!

This note is mainly used to record some trivial but interesting problems that I encounter in daily life.

For as the heavens are higher than the earth, so are my ways higher than your ways and my thoughts than your thoughts.—Isaiah 55:9



FIGURE 1. Night City

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¹This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre

1. 2023.12.22

1.1. Identification of co-character groups with Lie algebras. Settings: G complex connected reductive algebraic group. $Q \subseteq P \subseteq \mathfrak{h}^*$ are root lattice and weight group, respectively. $\check{Q} \subseteq \check{P} \subseteq \mathfrak{h}$ are coroot lattice and coweight group, respectively. H is a maximal torus of G , $\mathfrak{h}$ is its Lie algebra.

There are identifications

$$\begin{aligned} I: X_*(H) \otimes \mathbb{C} &\longrightarrow \mathfrak{h} \\ \phi &\longmapsto d\phi(1) \\ J: X^*(H) \otimes \mathbb{C} &\longrightarrow \mathfrak{h}^* \\ \varphi &\longmapsto d\varphi \end{aligned}$$

Under these identifications we have

$$\begin{aligned} \check{P} &= \{\check{\lambda} \in \mathfrak{h} \mid \langle \alpha, \check{\lambda} \rangle \in \mathbb{Z}, \text{ for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\langle \alpha, \check{\lambda} \rangle) = 1, \text{ for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \alpha(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{ for all } \alpha \in Q\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\check{\lambda}) \in Z(G)\}. \end{aligned}$$

And

$$\begin{aligned} X_*(H) &= \{\check{\lambda} \in \mathfrak{h} \mid d\phi(\check{\lambda}) \in \mathbb{Z}, \text{ for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}d\phi(\check{\lambda})) = 1, \text{ for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \phi(\exp(2\pi\sqrt{-1}\check{\lambda})) = 1, \text{ for all } \phi \in X^*(H)\} \\ &= \{\check{\lambda} \in \mathfrak{h} \mid \exp(2\pi\sqrt{-1}\check{\lambda}) = 1\}. \end{aligned}$$

Similarly

$$P = \{\lambda \in \mathfrak{h}^* \mid \exp(2\pi\sqrt{-1}\lambda) \in Z(\check{G})\}.$$

$$X^*(H) = \{\lambda \in \mathfrak{h}^* \mid \exp(2\pi\sqrt{-1}\lambda) = 1\}.$$

1.2. Pinnings of algebraic groups.

Theorem 1.1 (see AdC18). *Inner automorphism group $\text{Inn}(G)$ of G is equal to $G/Z(G)$, $\text{Inn}(G)$ acts freely and transitively on the set of pinnings.*

I know that there is an analogy in compact connected groups, and it has been proved.

Theorem 1.2. *If G is a compact connected group, then its inner automorphism group $\text{Inn}(G) = G/Z(G)$ acts freely and transitively on the set of pinnings. (where pinnings for it is a bit different from the complex case, that $(T, B_{\mathbb{C}}, \mathcal{X})$ is a pinning if T is a maximal torus, $B_{\mathbb{C}}$ a Borel of $G_{\mathbb{C}}$ containing T , $\mathcal{X}$ is a set of real rays in simple root space.)*

Proof. Here's proof from mathoverflow Lspise's answer:

To prove this, I'll use a few pieces of structure theory:

1. All maximal tori in G are G° -conjugate.
2. All Borel subgroups of $G_{\mathbb{C}}$ are $G_{\mathbb{C}}^\circ$ -conjugate.
3. For every maximal torus T in G , the map $W(G^\circ, T) \rightarrow W(G_{\mathbb{C}}^\circ, T_{\mathbb{C}})$ is an isomorphism.
4. If G_{sc} and $(G_{\mathbb{C}})_{sc}$ are the simply connected covers of the derived groups of G° and $G_{\mathbb{C}}^\circ$, then $(G_{sc})_{\mathbb{C}}$ equals $(G_{\mathbb{C}})_{sc}$.
5. [Every compact Lie group has a finite subgroup that meets every component.](#)

I only need (4) to prove that, for every maximal torus T in G , the map from T to the set of conjugation-fixed elements of $T/Z(G^\circ)$ is surjective. This is probably a well known fact in its own right for real-group theorists.

Now consider triples $(T, B_{\mathbb{C}}, \mathcal{X})$ as follows: T is a maximal torus in G ; $B_{\mathbb{C}}$ is a Borel subgroup of $G_{\mathbb{C}}^\circ$ containing $T_{\mathbb{C}}$, with a resulting set of simple roots $\Delta(B_{\mathbb{C}}, T_{\mathbb{C}})$; and $\mathcal{X}$ is a set consisting of a real ray in each complex simple root space (i.e., the set of positive real multiples of some fixed non-0 vector). (Sorry about the pair of modifiers "complex simple".) I will call these 'pinnings', although it doesn't agree with the usual terminology (where we pick individual root vectors, not rays). I claim that $G^\circ/Z(G^\circ)$ acts simply transitively on the set of pinnings.

Once we have transitivity, freeness is clear: if $g \in G^\circ$ stabilises some pair $(T, B_\mathbb{C})$, then it lies in T , and so stabilises every complex root space; but then, for it to stabilise some choice of rays $\mathcal{X}$, it has to have the property that $\alpha(g)$ is positive and real for each simple root α ; but also $\alpha(g)$ is a norm-1 complex number, hence trivial, for each simple root α , hence for each root α , so that g is central.

For transitivity, since (1) all maximal tori in G are G° -conjugate, so (2) for every maximal torus T in G , the Weyl group $W(G^\circ_\mathbb{C}, T_\mathbb{C})$ acts transitively on the Borel subgroups of $G^\circ_\mathbb{C}$ containing $T_\mathbb{C}$, and (3) $W(G^\circ, T) \rightarrow W(G^\circ_\mathbb{C}, T_\mathbb{C})$ is an isomorphism, it suffices to show that all possible sets $\mathcal{X}$ are conjugate. Here's the argument that I came up with to show that they are even T -conjugate; I think it can probably be made much less awkward. Fix a simple root α , and two non-0 elements X_α and X'_α of the corresponding root space. Then there are a positive real number r and a norm-1 complex number z such that $X'_\alpha = rzX_\alpha$. Choose a norm-1 complex number w such that $w^2 = z$. There is then a unique element s_{ad} of $T_\mathbb{C}/Z(G^\circ_\mathbb{C})$ such that $\alpha(s_{\text{ad}}) = w$, and $\beta(s_{\text{ad}}) = 1$ for all simple roots $\beta \neq \alpha$. By (4), we can choose a lift s_{sc} of s_{ad} to $(G_{\text{sc}})_\mathbb{C} = (G_\mathbb{C})_{\text{sc}}$, which necessarily lies in the preimage $(T_\mathbb{C})_{\text{sc}}$ of (the intersection with the derived subgroup of) T , and put $t_{\text{sc}} = s_{\text{sc}} \cdot \overline{s_{\text{sc}}}$. Then

$$\alpha(t_{\text{sc}}) = \alpha(s_{\text{sc}})\overline{\alpha(s_{\text{sc}})} = \alpha(s_{\text{sc}})\overline{\alpha(s_{\text{sc}})^{-1}} = w \cdot \overline{w^{-1}} = z,$$

and, similarly, $\beta(t_{\text{sc}}) = 1$ for all simple roots $\beta \neq \alpha$. Now the image t of t_{sc} in $G^\circ_\mathbb{C}$ lies in $T_\mathbb{C}$ and is fixed by conjugation, hence lies in T ; and $\text{Ad}(t)X_\alpha = zX_\alpha$ lies on the ray through X'_α .

Since G also acts on the set of pinnings, we have a well defined map $p : G \rightarrow G^\circ/Z(G^\circ)$ that restricts to the natural projection on G° . Now $\ker(p)$ meets every component, but it contains $Z(G^\circ)$, so it need not be finite. Applying (5) to the Lie group $\ker(p)$ yields the desired subgroup H . Note that, as requested in your [improved classification](#), conjugation by any element of H fixes a pinning, hence, if inner, must be trivial.

□

Remark 1.3. *The pinning maps show that split reductive algebraic groups look like butterflies!*

According to Milne, Grothendieck used the following analogy to talk about the structure theory of algebraic groups. An algebraic group is like a butterfly. The body of the butterfly is the maximal torus. The wings are two opposite Borel subgroups. And the pins are the pinning maps.



FIGURE 2. Pinning of butterfly

1.3. Coherent continuation representations. Until the end of today, G denotes a real reductive group in Harish-Chandra class.

Let's recall some basic definitions of coherent continuation representations, in order for me to do calculations in the case of real classical groups.

Fix a connected reductive complex Lie group $G_{\mathbb{C}}$, together with a Lie group homomorphism $\iota : G \rightarrow G_{\mathbb{C}}$ such that its differential $d\iota : \text{Lie}(G) \rightarrow \text{Lie}(G_{\mathbb{C}})$ has the following two properties:

- (1) the kernel of $d\iota$ is contained in the center of $\text{Lie}(G)$;
- (2) the image of $d\iota$ is a real form of $\text{Lie}(G)$.

The analytical weight lattice of $G_{\mathbb{C}}$ is identified with a subgroup of ${}^a\mathfrak{h}^*$ via $d\iota$. We write $Q_{\iota} \subset {}^a\mathfrak{h}^*$ for this subgroup. Denote root lattice by $Q_{\mathfrak{g}}$, weight lattice by $Q^{\mathfrak{g}}$, then

$$Q_{\mathfrak{g}} \subset Q_{\iota} \subset Q^{\mathfrak{g}}.$$

They are all W stable subgroups. And we have some categories:

- (1) $\text{Rep}(\mathfrak{g}, Q_{\iota})$ the category of all finite-dimensional representations of $\mathfrak{g}$ whose weight are contained in $Q_{\mathfrak{g}}$, **it can be identified with $\text{Rep}(\text{Lie}(G_{\mathbb{C}}), Q_{\iota})$ and hence $\text{Rep}_{\text{finite}}(G_{\mathbb{C}})$** ;
- (2) $\text{Rep}(G)$ the category of all Casselman-Wallach representations of G , and its full subcategories $\text{Rep}_{\mu}(G)$, $\text{Rep}_{\mathfrak{s}}(G)$.

To denote their Grothendieck groups, we replace Rep by $\mathcal{K}$.

Fix a Q_{ι} -coset $\Lambda = \lambda + Q_{\iota} \subset {}^a\mathfrak{h}^*$.

Definition 1.4 (coherent family). *A $\mathcal{K}(G)$ -valued Λ coherent family is a map*

$$\Phi : \Lambda \rightarrow \mathcal{K}(G)$$

satisfies the following two conditions:

- (1) *for all $\nu \in \Lambda$, $\Phi(\nu) \in \mathcal{K}_{\nu}(G)$;*
- (2) *for all representations $F \in \text{Rep}(\mathfrak{g}, Q_{\iota})$ and all $\nu \in \Lambda$,*

$$F \cdot (\Phi(\nu)) = \sum_{\mu} \Phi(\nu + \mu),$$

where μ runs over all weights of F , counted with multiplicities.

The set of coherent families is denoted by $\text{Coh}_{\Lambda}(\mathcal{K}(G))$

We have the integral Weyl group $W_{\Lambda} = \{w \in W \mid w\lambda - \lambda \in Q_{\iota}\}$ there is a representation of W_{Λ} on $\text{Coh}_{\Lambda}(\mathcal{K}(G))$ by

$$(w \cdot \Phi)(\nu) = \Phi(w^{-1}\nu).$$

This is called a coherent continuation representation.

2. 2023.12.23

2.1. Review of regular parameter. Following yesterday's discussion of coherent continuation representations, we define two basis for it, use Langland classification.

Suppose H is a Cartan subgroup of G , by which we mean a subgroup that is the centralizer of a Cartan subalgebra of $\text{Lie}(G)$ (In the case of linear groups, it's just a real algebraic torus.). H has a unique maximal compact subgroup (since G is in Harish-Chandra's class, adjoint representation of H is trivial, H_0 lies in the center of H , as we know, all maximal subgroups of Lie groups are conjugate to each other, use Cartan decomposition of H , we win. In fact, H has a unique maximal compact torus and a unique maximal split torus) We denote by $\Delta_{\mathfrak{h}} \subset \mathfrak{h}^*$ the root system of $\mathfrak{g}$. A root is called imaginary if $\check{\alpha} \in \mathfrak{t}$ (i.e. roots which take pure imaginary values on $\text{Lie}(H)$), an imaginary root α is called compact if $\mathfrak{g}_{\alpha}$ and $\mathfrak{g}_{-\alpha}$ is contained in the complexified Lie algebra of a common compact subgroup of G . (this is equivalent to $\mathfrak{g}_{\alpha}$ belong to a Lie subalgebra of a compact subgroup, since $\mathfrak{g}_{\alpha}$ lies in a Lie subalgebra of maximal compact subgroup is equivalent to $\mathfrak{g}_{-\alpha}$ Lies in the same subgroup)

There are two important facts about the representation theory of H :

- (1) Every Casselman-Wallach representation of H is finite-dimensional, it follows directly from that H is an extension of an abelian group and a finite group;
- (2) For every $\Gamma \in \text{Irr}(H)$ differential of Γ is a direct sum of one-dimensional representations attached to a unique $d\Gamma \in \mathfrak{h}^*$. (it is true because H is in Harish-Chandra's class).

For every Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ containing $\mathfrak{h}$, write

$$\xi_{\mathfrak{b}} : \mathfrak{h} \rightarrow {}^a\mathfrak{h},$$

for the linear isomorphism attached to $\mathfrak{b}$ defined by

$$\mathfrak{h} \hookrightarrow \mathfrak{b} \rightarrow \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}] = {}^a\mathfrak{h}.$$

The transpose inverse of this map is still denoted by $\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \rightarrow \mathfrak{h}^*$.

Write

$$W({}^a\mathfrak{h}^*, \mathfrak{h}^*) := \{\xi_{\mathfrak{b}} : {}^a\mathfrak{h}^* \rightarrow \mathfrak{h}^* \mid \mathfrak{b} \text{ is a Borel subalgebra containing } \mathfrak{h}\}.$$

Definition 2.1. For every element $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$, put

$$\delta(\xi) := \frac{1}{2} \cdot \sum_{\alpha \text{ is an imaginary root in } \xi\Delta^+} \alpha - \sum_{\beta \text{ is an compact imaginary root in } \xi\Delta^+} \beta \in \mathfrak{h}^*.$$

Write $\mathcal{P}_{\Lambda}(G)$ for the set of all triples $\gamma = (H, \xi, \Gamma)$, where H is a Cartan subgroup of G , $\xi \in W({}^a\mathfrak{h}^*, \mathfrak{h}^*)$, and

$$\begin{array}{ccc} \Gamma : & \Lambda & \longrightarrow \text{Irr}(H) \\ & \nu & \longmapsto \Gamma_{\nu} \end{array}$$

is a map with the following properties:

- (1) $\Gamma_{\nu+\beta} = \Gamma_{\nu} \otimes \xi(\beta)$ for all $\beta \in Q_{\iota}$ and $\nu \in \Lambda$;
- (2) $d\Gamma_{\nu} = \xi(\nu) + \delta(\xi)$ for all $\nu \in \Lambda$. (note that Cartan subgroups not always have irr repn with this differential!)

Here $\xi(\beta)$ is naturally viewed as a character of H by using the homomorphism $\iota : H \rightarrow H_{\mathbb{C}}$, and $H_{\mathbb{C}}$ is a Cartan subgroup of $G_{\mathbb{C}}$ containing $\iota(H)$.

2.2. Outer automorphism of algebraic groups.

Lemma 2.2 (a simple discover of abstract algebra). *G an abstract group, H a normal subgroup of G , if G acts transitively on a set X such that the restriction of this action to H is free and transitive, then the short exact sequence of groups*

$$1 \rightarrow H \rightarrow G \rightarrow G/H \rightarrow 1$$

Proof. Actually choose an element $x \in X$, we have

$$G \simeq \text{Stab}_G(x) \ltimes H.$$

□

Using this lemma, we can see that for a complex connected reductive algebraic group G , $\text{Aut}(G)$ acts on the set of pinning transitivity, with its normal subgroup the $\text{Inn}(G)$ acts freely and transitively, so every pinning gives us a split of the short exact sequence

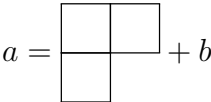
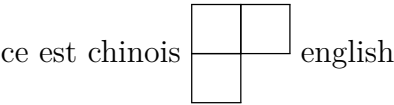
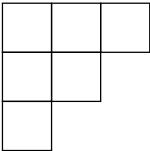
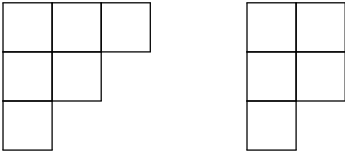
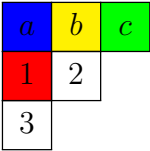
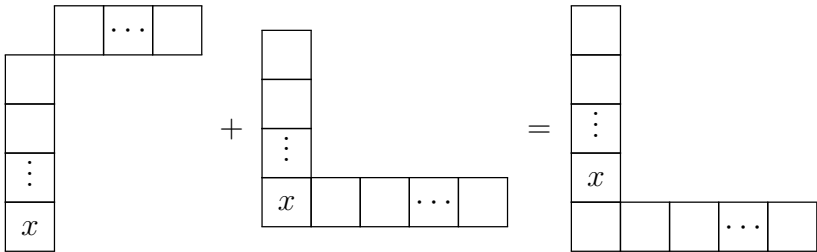
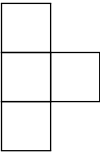
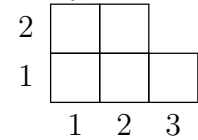
$$1 \rightarrow \text{Inn}(G) \rightarrow \text{Aut}(G) \rightarrow \text{Out}(G) \rightarrow 1.$$

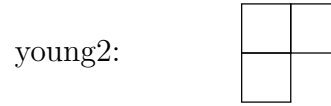
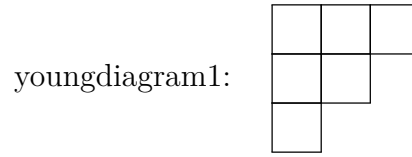
Some facts about split connected reductive algebraic groups over arbitrary field k :

- (1) $\{\text{split connected reductive algebraic groups}\}/\sim = \{\text{root datums}\}/\sim$;
- (2) For any two automorphisms of G , if they induce the same automorphism of root datum, then they differ by an inner automorphism;
- (3) Fix a pinning, we get an isomorphism $\text{Out}(G) \simeq \text{Aut}(D_b)$.

3. 2024.2.27

Today, I learned how to draw the Young tableau.





For a compact connected Lie group G , choose a maximal torus T , there is a canonical 1-1 correspondence:

$$X^*(T)/W \longleftrightarrow (X^*(T) + \rho)/W \longleftrightarrow \text{Irr}(G)$$

Moreover, if we choose a positive system Ψ^+ , and then ρ_{Ψ^+} , the above sets also canonically isomorphic to the set of dominant analytically integral weights:

$$\{\lambda \in X^*(T) \mid \langle \lambda, \alpha \rangle \geq 0, \text{ for all } \alpha \in \Psi^+\}$$

This can be generalized to parameter of discrete series representations for connected semisimple Lie groups G with compact Cartan subgroup T :

$$(X^*(T) + \rho)^{reg}/W_c$$

If a discrete series representation V has minimal K type which has maximal weight ν , then it's parameter is $\lambda = \nu - \rho + 2\rho_c$. (not sure)

Note that for a fixed infinitesimal character, there are exactly $|W/W_c|$ different discrete series representations.

4. 2024.2.28

Some facts about continuous complex characters of some common abelian topological groups:

field	character group	correspondence
$(\mathbb{R}_+^\times, \times)$	$\mathbb{C}$	$\xi \mapsto (t \mapsto t^\xi)$
$(\mathbb{C}^\times, \times)$	$\mathbb{C} \times \mathbb{Z}$	$(\xi, n) \mapsto (z \mapsto z ^\xi (\frac{z}{ z })^n)$
$(\mathbb{S}^1, \times)$	$\mathbb{Z}$	$n \mapsto (z \mapsto z^n)$
$(\mathbb{Q}_p, +)$	$\mathbb{Q}_p$	$\xi \mapsto (x \mapsto \exp(2\pi i \xi \{x\}_p))$

Remark 4.1. *Note that all continuous characters of $\mathbb{Q}_p$ are automatically locally constant (smooth), which is consistent with the case of real lie groups.*

5. 2024.10.22

Lemma 5.1. *The associated variety of an irreducible Casselman-Wallach representation of a real reductive group in Harish-Chandra class is the closure of a nilpotent orbit in $\mathfrak{g}^*$.*

Proof. By the work of Borho, Brylinski [BB], and Joseph [Jos], the associated variety of any primitive ideal (annihilator of an irreducible $\mathfrak{g}$ -module) is the closure of a single nilpotent orbit. So we only need to prove that $\text{gr}(\text{Ann}_{\mathcal{U}(\mathfrak{g})}(V))$ is a primitive ideal.

Since the K -finite part V_K is dense in V , $\text{Ann}_{\mathcal{U}(\mathfrak{g})}(V) = \text{Ann}_{\mathcal{U}(\mathfrak{g})}(V_K)$, we only need to consider the annihilator of the irreducible $(\mathfrak{g}, K)$ -module V_K .

If G is connected, then K is also connected, so V_K is irreducible as $\mathcal{U}(\mathfrak{g})$ -module, in this case, $\text{Ann}_{\mathcal{U}(\mathfrak{g})}(V)$ is a primitive ideal.

If G is not connected, let G_0 be the identity component of G , $K_0 = G_0 \cap K$ is a maximal compact subgroup of G_0 , then $V_K|_{G_0}$ is a finite length $(\mathfrak{g}, K_0)$ -module, it has an irreducible quotient $V_K \rightarrow V_0$, then according to the theory of induction of $(\mathfrak{g}, K)$ -modules [KV, Chapter 2] we have an injective morphism of $(\mathfrak{g}, K_0)$ -modules:

$$(5.1) \quad V_K|_{K_0} \hookrightarrow (\text{induced}_{K_0}^K(V_0))|_{K_0} \cong \bigoplus_{\text{double cosets } K_0 k K_0} kV_0,$$

where kV_0 be the $(\mathfrak{g}, K_0)$ -module whose underlying space is V_0 , whose action by $X \in \mathfrak{g}$ is the usual action by $\text{Ad}(k)X$, whose action by $g \in K_0$ is the usual action by $k^{-1}gk$, it has the same annihilator in $\mathcal{U}(\mathfrak{g})$ as V_0 since G is in Harish-Chandra's class, and hence the annihilator of V_K is the same as the annihilator of V_0 , which is a primitive ideal.

□

6. 2024.11.28

Any split connected reductive group over arbitrary field can be uniquely defined and split over $\mathbb{Z}$. (Chevalley group)

Classification of split reductive group is the same over arbitrary non-empty schemes, which is equivalent to the classification of root data.(cf. [SGA3])

7. 2024.12.17

7.1. Geometric Pre-quantization. Let X be a smooth manifold. There are natural maps

$$\mathrm{Pic}(X) \longrightarrow H_{dR}^2(X; \mathbb{R}) \simeq H_{Cech}^2(X; \mathbb{R}).$$

Construction of the first map: For a (complex) line bundle $\mathbb{L} \in \mathrm{Pic}^h(X)$, let $\langle, \rangle$ be a Hermitian metric on it and ∇ be a unitary connection. (Every line bundle $\mathbb{L} \in \mathrm{Pic}(X)$ has a Hermitian metric and a unitary connection on it.)

- choose a locally trivialization $\{U_\alpha\}$ of $\mathbb{L}$, such that all U_α and $U_\alpha \cap U_\beta$ are contractible;
- choose Unitary local frames $\{s_\alpha\}$ for $\mathbb{L}$ on each U_α , then $s \leftrightarrow (f_\alpha)$ and $\nabla \leftrightarrow (\theta_\alpha)$ (unitary implies that θ_α are pure imaginary);
- define $\Omega := d\theta_\alpha$, and $c_1(\mathbb{L}) := [\frac{1}{2\pi i}\Omega] \in H_{dR}^2(X; \mathbb{R})$ ($\Omega = \mathrm{curv}\nabla$).

$c_1(\mathbb{L})$ is independent of the choice of $\langle, \rangle$ and ∇ by Chern-Weil theory. And we can see that $\theta_\alpha - \theta_\beta = \frac{dg_{\alpha,\beta}}{g_{\alpha,\beta}} = d\log(g_{\alpha,\beta})$ on $U_\alpha \cap U_\beta$, where $g_{\alpha,\beta}$ is the transition function ($s_\beta = g_{\alpha,\beta}s_\alpha$).

Construction of the second map (classical de-Rham isomorphism), choose a 2-form $\omega \in H_{dR}^2(X; \mathbb{R})$

- choose a cover $\{U_\alpha\}$ of $\mathbb{L}$, such that all U_α and $U_\alpha \cap U_\beta$ are contractible;
- on each U_α , choose a 1-form θ_α such that $d\theta_\alpha = \omega|_{U_\alpha}$;
- since $d\theta_\alpha = d\theta_\beta$ on $U_\alpha \cap U_\beta$, we can choose smooth function $c_{\alpha,\beta}$ such that $dc_{\alpha,\beta} = \theta_\alpha - \theta_\beta$;
- let $c_{\alpha,\beta,\gamma} = c_{\alpha,\beta} + c_{\beta,\gamma} + c_{\gamma,\alpha}$, which is a constant, and we get the corresponding 2-cocycle $(c_{\alpha,\beta,\gamma}) \in H_{Cech}^2(X; \mathbb{R})$.

If ω come from the first map, we can choose $c_{\alpha,\beta} = \frac{1}{2\pi i} \log g_{\alpha,\beta}$, and hence $c_{\alpha,\beta,\gamma} = \frac{1}{2\pi i} (\log g_{\alpha,\beta} + \log g_{\beta,\gamma} + \log g_{\gamma,\alpha}) \in \mathbb{Z}$. So the morphism is in fact

$$\mathrm{Pic}(X) \longrightarrow H_{dR}^2(X; \mathbb{Z}) \simeq H_{Cech}^2(X; \mathbb{Z}).$$

In fact the first map is also an isomorphism. (cf. prequantization) For a 2-form ω , choose θ_α and $c_{\alpha,\beta}$ as above, let $g_{\alpha,\beta} = \exp(2\pi i c_{\alpha,\beta})$, these are transition functions of a complex line bundle $\mathbb{L}$.

Corollary 7.1. *Assume that G is a connected Lie group. The following are equivalent:*

- $\Omega \subseteq \mathfrak{g}^*$ is integral ($[\sigma_\Omega] \in H^2(X; \mathbb{Z})$);
- there exists a G -equivariant complex line bundle over Ω with a G -invariant Hermitian connection ∇ such that

$$\frac{1}{2\pi i} \mathrm{curv}(\nabla) = \sigma_\Omega.$$

Theorem 2.6 (Weil). *Let M be a smooth manifold and ω a real, closed 2-form whose cohomology class $[c]$ is integral. Then there is a unique Hermitian line bundle $\mathbb{L}$ over M with unitary connection ∇ so that $c_1(\mathbb{L}) = [c]$.*

Sketch of Proof. Existence: Reverse the arguments above. Since $[c]$ is integral, $e^{2\pi i c_{ijk}} = 1$. As a consequence, $g_{ij}g_{jk}g_{kl} = 1$. So g_{ij} 's are the transition function for some line bundle whose first Chern class is $[c]$.

Uniqueness: Suppose $c_1(\mathbb{L}) = c_1(\tilde{\mathbb{L}})$. Let $h_{ij} = \frac{1}{2\pi i} g_{ij}$ and define $\tilde{h}_{ij}$ similarly. Then the functions $\hat{h}_{ij} = h_{ij} - \tilde{h}_{ij}$ satisfies the relation

$$\hat{h}_{ij} + \hat{h}_{jk} + \hat{h}_{ki} = 0.$$

We take a partition of unity ρ_k and let $\lambda_i = e^{2\pi i \sum \hat{h}_{ki} \rho_k}$. Then

$$\lambda_i g_{ij} \lambda_j^{-1} = e^{2\pi i (h_{ij} + \sum (\hat{h}_{ki} - \hat{h}_{kj}) \rho_k)} = e^{2\pi i (h_{ij} - \hat{h}_{ij})} = \tilde{g}_{ij}.$$

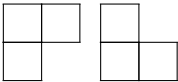
So as line bundles $\mathbb{L} \simeq \tilde{\mathbb{L}}$. □

FIGURE 3. Proof

8. 2025.4.7

TooYoung package from xionggrui.

1	$i + 1$
2	
3	
...	
i	
$i + 2$	
...	
n	



9. 2025.8.14 ²

Let G be a finite central extension of an open subgroup of $\mathbb{R}$ -points of a connected reductive algebraic group (reductive Nash group [S]) with Lie algebra $\mathfrak{g}$. Fix an invariant nondegenerate bilinear form $B(-, -)$ on $\mathfrak{g}$.

For an irreducible Casselman-Wallach representation π of G , let $\text{Wh}_{\mathcal{O}}(\pi)$ be the space of generalized Whittaker models of π associated to the nilpotent orbit $\mathcal{O}$. Let

$$\text{WO}(\pi) := \{\mathcal{O} \mid \mathcal{O} \text{ is a nilpotent orbit such that } \text{Wh}_{\mathcal{O}}(\pi) \neq 0\},$$

and the *Whittaker support* $\text{WS}(\pi)$ of π be the set of all maximal orbits in $\text{WO}(\pi)$ with respect to closure inclusion:

$$\text{WS}(\pi) := \{\mathcal{O} \mid \mathcal{O} \text{ is a maximal orbit in } \text{WO}(\pi)\}.$$

Another nilpotent invariant is the wavefront cycle:

$$\mathcal{WF}(\pi) = \sum_i m_i [\mathcal{O}_i].$$

With the corresponding wavefront set :

$$\text{WF}(\pi) = \bigcup_{m_i \neq 0} \overline{\mathcal{O}_i}.$$

Mœglin-Waldspurger [MW87] proved that $\text{WF}(\pi) = \text{WS}(\pi)$ when F is a p -adic field. We want to test this conjecture for arbitrary reductive almost linear Nash groups.

Conjecture 9.1 (Main conjecture). *For any reductive almost linear Nash group G and any irreducible Casselman-Wallach representation π :*

$$\text{WF}(\pi) = \text{WS}(\pi).$$

Take a nilpotent orbit $\mathcal{O} \subseteq \mathfrak{g}$ (or $\mathfrak{sl}_2$ -triple $\gamma = \{Y, H, X\}$ in $\mathfrak{g}$). $\mathfrak{g}/\mathfrak{g}^Y$ is a symplectic space with symplectic form $\langle -, - \rangle := B(Y, [-, -])$. Set $G^Y := \text{Stab}_G(Y)$, there is a natural homomorphism $G^Y \rightarrow \text{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -, - \rangle)$. Let $\widetilde{G}^Y$ be the double cover of G^Y induced by the metaplectic double cover of $\text{Sp}(\mathfrak{g}/\mathfrak{g}^Y, \langle -, - \rangle)$.

Definition 9.2. *The nilpotent $\mathcal{O}$ is called admissible if the double cover $\widetilde{G}^Y$ is split over the identity component of $(G^Y)_0$. It is called quasi-admissible if $\widetilde{G}^Y$ has a finite-dimensional genuine representation.*

Result of [GGS21] says that $\text{WS}(\pi)$ consists of quasi-admissible orbits. But they do not know whether the non-admissible quasi-admissible orbits appear in Whittaker supports of representations. We may ask the same question for the wavefront set:

Problem 1: Are all the nilpotent orbits in the wavefront cycle admissible?

For the case when G is general linear, it is well known that the wavefront set $\text{WF}(\pi)$ of an irreducible Casselman-Wallach representation π of G consists of a single nilpotent adjoint orbit. Together with some counting results, it maybe possible to construct all irreducible representations of general linear group via theta correspondence and coherent continuation.

Problem 2: Can we construct all irreducible representations via theta correspondence, coherent continuation, and twist of characters?

²This section is organized by Ryan Ang and Qiutong Wang at Tianyuan Mathematics Research Centre

Assuming the construction steps above are done, it then remains to study the Whittaker support $\text{WS}(\pi)$ of π . We focus on two types of interactions with Whittaker models - theta correspondence and coherent continuation.

On one hand, Gomez-Zhu [GZ14] proved for type I reductive (symplectic-orthogonal) dual pairs in the stable range that the big theta lift preserves the structure of Whittaker models:

$$\text{Wh}_{\Theta(\mathcal{O})}(\Theta(\pi)) = \text{Wh}_{\mathcal{O}}(\pi^\vee).$$

Noting that $(G, G') = (\text{GL}_n(F), \text{GL}_m(F))$ is a (reductive) dual pair for any $n, m \in \mathbb{N}$, one observes that if one can prove an analogue of Gomez-Zhu's result for type II dual pairs, one can use it to infer properties of the structure of the Whittaker support $\text{WS}(\pi)$. Hence we propose the following subconjecture:

Conjecture 9.3. *If $(G, G') = (\text{GL}_n(F), \text{GL}_m(F))$ is a type II reductive dual pair over F in the stable range (with $n \leq m$), and π is a genuine Casselman-Wallach representation of G , then*

$$\text{Wh}_{\Theta(\mathcal{O})}(\Theta(\pi)) = \text{Wh}_{\mathcal{O}}(\pi^\vee),$$

where $\pi^\vee$ denotes the smooth contragredient of π . Furthermore, the above will imply that

$$\text{WS}(\Theta(\pi)) = \text{WS}(\pi^\vee).$$

On the other hand, Barbash-Vogan [BV] proved that under coherent continuation, the multiplicity of wavefront cycle changes in the form of a Goldie rank polynomial, i.e.

$$\mathcal{WF}(\pi(\lambda)) = p(\lambda) \cdot \left(\sum_i c_i [\mathcal{O}_i] \right),$$

where $p(\lambda)$ is the goldie rank polynomial, λ is dominant. We expect some similar result for the multiplicity of the generalized Whittaker model.

Problem 3: For a fixed nilpotent orbit $\mathcal{O}$, does there exist a constant $c_{\mathcal{O}}$ such that $\dim \text{Wh}_{\mathcal{O}}(\pi(\lambda)) = c_{\mathcal{O}} \cdot p(\lambda)$?

10. 2025.10.1

Orbit method for nilpotent Lie groups.

Let G be a connected, simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$. Let $\mathfrak{g}^*$ be the dual space of $\mathfrak{g}$.

We have fourier transform:

$$\mathcal{F} : \mathcal{S}(\mathfrak{g})dX \rightarrow \mathcal{S}(\sqrt{-1}\mathfrak{g}^*), \quad fdX \mapsto \widehat{fdX}, \quad \widehat{fdX}(l) = \int_{\mathfrak{g}} f(X)e^{\langle l, X \rangle} dX.$$

Take its dual:

$$\mathcal{S}'(\sqrt{-1}\mathfrak{g}^*) \rightarrow (\mathcal{S}(\mathfrak{g})dX)', \quad T \mapsto \check{T}, \quad \langle \check{T}, \widehat{fdX} \rangle = \langle T, fdX \rangle.$$

For any irreducible unitary representation π of G , we have its character $\Theta_\pi \in C^{-\infty}(G)$ defined by

$$\Theta_\pi(f) = \text{Tr}(\pi(fdg)), \quad fdg \in C_c^\infty(G)dg.$$

Using the exponential map $\exp : \mathfrak{g} \rightarrow G$ (isomorphism in this case), we can view Θ_π as a distribution on $\mathfrak{g}$:

$$\theta_\pi(fdX) = \Theta_\pi(f \circ \exp), \quad fdX \in C_c^\infty(\mathfrak{g})dX.$$

In fact, θ_π is a tempered generalized function on $\mathfrak{g}$ (it can continuously extends to $\mathcal{S}(\mathfrak{g})dX$). By the work of Howe, Kirillov, and others, we have the following theorem:

Theorem 10.1. *There exists a unique coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$ such that $\check{\theta}_\pi = \mu_{\mathcal{O}}$, where $\mu_{\mathcal{O}}$ is the canonical measure on $\mathcal{O}$.*

This defines a bijection between $\widehat{G}$ (the unitary dual of G) and the set of coadjoint orbits in $\sqrt{-1}\mathfrak{g}^*$.

Conversely, for any coadjoint orbit $\mathcal{O} \subseteq \sqrt{-1}\mathfrak{g}^*$, we can construct an irreducible unitary representation $\pi_{\mathcal{O}}$ of G such that $\check{\theta}_{\pi_{\mathcal{O}}} = \mu_{\mathcal{O}}$. This is the more interesting part, use the method of geometric quantization.

Take any element $\lambda \in \mathcal{O}$, we have an identification $\mathcal{O} = G/G_\lambda$, $\lambda : \mathfrak{g}_\lambda \rightarrow \mathbb{C}$ is a Lie algebra homomorphism, and hence $\chi_\lambda : G_\lambda \rightarrow \mathbb{C}^\times$ defined by $\chi_\lambda(\exp X) = e^{\langle \lambda, X \rangle}$ is a unitary character of G_λ . Then we can defined a Hermitian line bundle $\mathcal{L}_\lambda = G \times^{G_\lambda} \mathbb{C}$ over $\mathcal{O}$, with a natural G -invariant connection ∇ such that $\text{curv}(\nabla) = \omega_{\mathcal{O}}$, where $\omega_{\mathcal{O}}$ is the Kirillov-Kostant-Souriau symplectic form on $\mathcal{O}$.

Then we can define the Hilbert space of square integrable sections of $\mathcal{L}_\lambda$, but this space is too large. We need to choose a polarization $\mathfrak{h} \subseteq \mathfrak{g}_{\mathbb{C}}$ (maximal isotropic subalgebra with respect to the bilinear form $(X, Y) \mapsto \langle \lambda, [X, Y] \rangle$) and consider the space of square integrable holomorphic sections of $\mathcal{L}_\lambda$ which are covariant constant along $\mathfrak{h}$.

In fact the polarization $\mathfrak{h}$ corresponds to a G -invariant integrable distribution on $\mathcal{O}$, its a subbundle P of $T\mathcal{O}$ each fiber is a lagrangian subspace. Then we can define the space of polarized sections:

$$\Gamma_P(\mathcal{O}, \mathcal{L}_\lambda) = \{s \in \Gamma^\infty(\mathcal{O}, \mathcal{L}_\lambda \otimes |\omega_{\mathcal{O}/P}|^{\frac{1}{2}}) \mid \nabla_X s = 0, \forall X \in P\}.$$

On this space, we have a unitary G action and get a unitary representation $\pi_{\mathcal{O}}$ of G , it turns out that $\pi_{\mathcal{O}}$ is irreducible and independent of the choice of the polarization $\mathfrak{h}$.

Kirillov's philosophy works well for nilpotent Lie groups, but for more general Lie groups, it is not true anymore.

Now assume G is a real reductive group, with Lie algebra $\mathfrak{g}_0$. For any irreducible Casselman-Wallach representation π of G , we still have its generalized function character $\Theta_\pi \in C^{-\infty}(G)$, and hence a tempered generalized function θ_π on $\mathfrak{g}$. Here $\theta_\pi \in C^{-\infty}(\mathfrak{g}_0)$ is defined as follows:

$$\theta_\pi(f dX) = \Theta_\pi(\exp_*(\sqrt{\det(\exp)} f dX)), \quad f dX \in C_c^\infty(\mathfrak{g}_0) dX.$$

The generalized function Θ_π has the following properties:

- Θ_π is invariant under conjugation of G ;
- Θ_π is an eigendistribution of $\mathcal{U}(\mathfrak{g})^G$;
- Θ_π is locally integrable, and real analytic on the regular semisimple locus G^{rs} .

By the work of Rossmann, we have the following theorem:

Theorem 10.2. *If π is tempered, then*

$$\theta_\pi|_{\text{nbhd of } 0} = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}|_{\text{nbhd of } 0}.$$

Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$, is the orbital integral along $\mathcal{O}_i$.

For an arbitrary generalized function $\Theta \in C^{-\infty}(M)$, we can define its wavefront set $\text{WF}(\Theta) \subseteq \sqrt{-1}\text{T}^*M$ as follows:

Definition 10.3. *For any $(x, \xi) \in \sqrt{-1}\text{T}^*M$, $(x, \xi) \notin \text{WF}(\Theta)$ if there exists a compactly supported smooth function φ on M such that $\varphi(x) \neq 0$, and an open conic neighborhood Γ of ξ such that for any $N \in \mathbb{N}$, there exists a constant $C_N > 0$ such that*

$$|\widehat{\varphi\Theta}(t\eta)| \leq C_N t^{-N}, \quad \forall t > 0, \eta \in \Gamma.$$

$\text{WF}(\Theta)$ is a closed conic subset of $\sqrt{-1}\text{T}^*M$ (singular direction of Θ).

In particular, we can define the wavefront set $\text{WF}(\pi)$ of an irreducible Casselman-Wallach representation π of G as $\text{WF}(\Theta_\pi) \cap (\sqrt{-1}\text{T}_1^*(G))$, it is a closed conic G -invariant subset of $\sqrt{-1}\mathfrak{g}_0^*$. The GK-dimension of π is defined as

$$\text{Dim}(\pi) = \frac{1}{2} \dim(\text{WF}(\pi)).$$

θ_π is a real analytical function on $\mathfrak{g}_0^{\text{rs}}$, and

$$\theta_\pi^*(X) := \lim_{t \rightarrow 0} t^{\text{Dim}(\pi)} \theta_\pi(tX)$$

is still a real analytical function on $\mathfrak{g}_0^{\text{rs}}$, where $\mathcal{B}$ is the flag variety of G . We can consider it as in $C^{-\infty}(\mathfrak{g}_0)$.

Theorem 10.4. *(Barbasch-Vogan) The asymptotic of θ_π defined above is a non-negative linear combination of Fourier transforms of nilpotent orbital integrals:*

$$\theta_\pi^* = \sum_i c_i \widehat{\mu_{\mathcal{O}_i}}.$$

Here $\mu_{\mathcal{O}_i} \in \mathcal{S}(\sqrt{-1}\mathfrak{g}_0^*)$, is the orbital integral along $\mathcal{O}_i$.

$$\text{WF}(\pi) = \bigcup_{c_i \neq 0} \overline{\mathcal{O}_i}.$$

11. 2025.10.2

11.1. Conormal bundle of singular subvariety. Let $\mathcal{B}$ be the flag variety of G . $\overline{\mathcal{B}}_y \subsetneq \overline{\mathcal{B}}_w$ two schubert varieties, what is the relation between their conormal bundles $T_{\mathcal{B}_y}^* \mathcal{B}$ and $T_{\mathcal{B}_w}^* \mathcal{B}$?

Let X be a smooth algebraic variety over $\mathbb{C}$, $Z \subseteq X$ be a closed subvariety, define the conormal bundle $T_Z^* X$ of Z in X as follows:

$$T_Z^* X = \overline{T_{Z^{\text{reg}}}^* X}.$$

For a singular point $z \in Z$ what is the fiber of $T_Z^* X$ at z ?

Here is a simple example:

Example 11.1. Let $X = \mathbb{C}^2$ and

$$Z = \{xy = 0\} \subset X$$

be the union of the coordinate axes.

- The regular locus is

$$Z^{\text{reg}} = (x\text{-axis} \setminus \{0\}) \cup (y\text{-axis} \setminus \{0\}).$$

- The singular locus is

$$S := Z \setminus Z^{\text{reg}} = \{0\}.$$

Now, the fibre of the conormal bundle at the origin is

$$(T_Z^* X)_0 = \text{Ann}(T_{x\text{-axis}}) \cup \text{Ann}(T_{y\text{-axis}}) = \langle dy \rangle \cup \langle dx \rangle,$$

which is a union of two lines in $T_0^* \mathbb{C}^2$.

On the other hand,

$$(T_S^* X)_0 = (T_{\{0\}}^* X)_0 = T_0^* X,$$

the entire cotangent plane.

Thus,

$$(T_Z^* X)_0 \subsetneq (T_{Z-Z^{\text{reg}}}^* X)_0,$$

11.2. Summary of paper [KT, The characteristic cycles of holonomic systems on a flag manifold]. Let G be a semisimple simply connected complex algebraic group. Define Steinberg variety:

$$Z = \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \mid x \in \text{Lie}(B') \cap \text{Lie}(B'')\} (= \{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \mid \}).$$

(= $\{(x, B', B'') \in \mathcal{N} \times \mathcal{B} \times \mathcal{B} \mid x \in \mathfrak{n}^\perp \cap \mathfrak{n}''^\perp\}$ more canonically.)

Here $\mathcal{N} \subseteq \mathfrak{g}$ is the nilpotent cone of $\mathfrak{g}$, $\mathcal{B}$ is the flag variety of G . It can be identify with the union of conormal bundles of all G -orbits in $\mathcal{B} \times \mathcal{B}$:

$$Z = \bigcup_{w \in W} T_{\mathcal{Y}_w}^* (\mathcal{B} \times \mathcal{B}),$$

which is a closed conic lagrangian subvariety of $T^*(\mathcal{B} \times \mathcal{B})$.

We also have the moment map:

$$\mu : Z \rightarrow \mathcal{N}, \quad (x, B', B'') \mapsto x.$$

By taking characteristic cycles, we get a map from the (complex) Grothendieck group of G -equivariant holonomic D -modules on $\mathcal{B} \times \mathcal{B}$ to the top Borel-Moore homology of Z :

$$\text{CC} : K_0(\text{Mod}_{\Delta G}^{\text{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \rightarrow H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}).$$

We also have isomorphisms:

$$h : K_0(\text{Mod}_{\Delta G}^{\text{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \rightarrow \mathbb{C}[W], \quad \mathcal{M}_w \mapsto w, \quad \mathcal{L}_w \mapsto a(w)$$

By Kazhdan-Lusztig, we have:

$$a(w) = \sum_{y \leq w} (-1)^{l(y)+l(w)} P_{y,w}(1) y.$$

There is a convolution product on $K_0(\text{Mod}_{\Delta G}^{\text{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}}))$ which makes h an isomorphism of algebras.

On

$$H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}) \simeq \bigoplus_{w \in W} H_{\text{top}}^{\text{BM}}(\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}, \mathbb{C}) \simeq \bigoplus_{w \in W} \mathbb{C}[\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}],$$

In the paper of Kazhdan-Lusztig [KL], they use topological method to construct a $W \times W$ action on $H_{\text{top}}^{\text{BM}}(Z, \mathbb{C})$.

Moreover there is a $W \times W$ module decomposition:

$$H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}),$$

where $\mathcal{O}$ runs over all nilpotent orbits in $\mathfrak{g}^*$, $Z_{\mathcal{O}} = \mu^{-1}(\mathcal{O})$. There is also an isomorphism of $W \times W$ -modules:

$$H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq (H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C}) \otimes H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C}))_{A_G(x)},$$

($A_G(x)$ coinvariants) where $x \in \mathcal{O}$, $\mathcal{B}_x = \{B' \in \mathcal{B} \mid x \in \text{Lie}(B')\}$ (the Springer fiber), $A_G(x) = \text{Stab}_G(x)/\text{Stab}_G(x)^0$.

By classical Springer theory, $H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C})$ is a representation of $W \times A_G(x)$, and we have the following decomposition:

$$H_{\text{top}}^{\text{BM}}(\mathcal{B}_x, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \rho.$$

Here $\tau(x, \rho)$ is an irreducible representation of W or zero.

So we have a decomposition of $W \times W$ -modules:

$$H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \tau(x, \rho).$$

Combine the above together, we finally have the following decomposition:

$$H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}) \simeq \bigoplus_{\rho \in \widehat{A_G(x)}} \tau(x, \rho) \otimes \tau(x, \rho), \quad H_{\text{top}}^{\text{BM}}(Z, \mathbb{C}) \simeq \bigoplus_{\mathcal{O}} H_{\text{top}}^{\text{BM}}(Z_{\mathcal{O}}, \mathbb{C}).$$

For any $w \in W$, there is a unique nilpotent orbit $\text{St}(w)$ such that $\mu(\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}) = \overline{\text{St}(w)}$. The map $w \mapsto \text{St}(w)$ defines a **surjective(?)** map from W to the set of nilpotent orbits in $\mathfrak{g}^*$ (this map can be computed via Robinson-Schensted algorithm).

Corollary 11.2. *For any nilpotent orbit $\mathcal{O}$, there is an isomorphism of $W \times W$ -modules:*

$$\bigoplus_{w \in W, \text{St}(w) = \mathcal{O}} \mathbb{C}[\overline{T_{\mathcal{Y}_w}^*(\mathcal{B} \times \mathcal{B})}] \simeq \bigoplus_{\rho \in \widehat{A_G(\mathcal{O})}} \tau(\mathcal{O}, \rho) \otimes \tau(\mathcal{O}, \rho).$$

Theorem 11.3. (*Main theorem of [KT]*) *The map*

$$\mathrm{CC} : K_0(\mathrm{Mod}_{\Delta G}^{\mathrm{Coh}}(\mathcal{D}_{\mathcal{B} \times \mathcal{B}})) \rightarrow H_{\mathrm{top}}^{\mathrm{BM}}(Z, \mathbb{C}),$$

is an isomorphism and $W \times W$ -equivariant.

Define $b(w) = h \circ \mathrm{CC}^{-1}([\overline{T_{y_w}^* \mathcal{B} \times \mathcal{B}}])$. Now we have three basis of $\mathbb{C}[W]$:

$$\{w\}_{w \in W}, \quad \{a(w)\}_{w \in W}, \quad \{b(w)\}_{w \in W}.$$

It was conjectured by Kashiwara and Tanisaki that: $a(w) = b(w)$ when $G = \mathrm{SL}_n(\mathbb{C})$ which means the characteristic cycle of $\mathcal{L}_w$ is irreducible, but unfortunately it is not true. Counterexample exist when $G = \mathrm{SL}_8(\mathbb{C})$, where Kashwara-Saito singularities appears at some Schubert varieties.

12. 2025.10.5

The Springer representation

Let G be a connected complex reductive group, K a symmetric subgroup of G . We can define the Springer representation of W on $H_c^\bullet(T_K^* \mathcal{B}, \mathbb{Q})$.

Recall that we have the following diagram:

$$\begin{array}{ccccc} \widetilde{\mathcal{N}} & \longrightarrow & \widetilde{\mathfrak{g}} & \longleftarrow & \widetilde{\mathfrak{g}}^{\mathrm{rs}} \\ \downarrow p_{\mathcal{N}} & & \downarrow p & & \downarrow p_{\mathrm{rs}} \\ \mathcal{N} & \longrightarrow & \mathfrak{g} & \longleftarrow & \mathfrak{g}^{\mathrm{rs}} \end{array}$$

In this diagram p_{rs} is a W -torsor. So we have a natural W -action on $p_{\mathrm{rs},*} \underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}^{\mathrm{rs}}}$ and hence on $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}}, \mathcal{L})$, where $\mathcal{L} = (p_{\mathrm{rs}})_* \underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}^{\mathrm{rs}}}$. There is a canonical isomorphism of perverse sheaves $\mathrm{IC}(\mathfrak{g}^{\mathrm{rs}}, \mathcal{L}) \simeq \mathbb{R}p_* \underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[\dim(\mathfrak{g})]$, so there is a natural W -action on $\mathbb{R}p_* \underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}$. By proper base change, we have

$$\mathbb{R}p_{\mathcal{N},*} \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}} \simeq \mathbb{R}p_* \underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}|_{\mathcal{N}}. \quad (\text{Spr} = \mathbb{R}p_{\mathcal{N},*} \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}[\dim \mathcal{N}] \text{ called } \textit{The Springer sheaf})$$

So there is a natural W -action on $\mathbb{R}p_{\mathcal{N},*} \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}$ (actually $\mathrm{End}_{\mathrm{Perv}(\mathcal{N})}(\mathrm{Spr}) = \mathbb{Q}[W]$) and hence on $H_c^\bullet(T_K^* \mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathcal{N}, \mathbb{R}p_{\mathcal{N},*} \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$ and $H_c^\bullet(\mathcal{B}_x, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\{x\}, \mathbb{R}p_{\mathcal{N},*} \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})$. (action on Springer fiber)

Since $T_K^* \mathcal{B} = p_{\mathcal{N}}^{-1}(\mathfrak{p})$, we also have a natural W -action on

$$H_c^\bullet(T_K^* \mathcal{B}, \mathbb{Q}) \simeq \mathbb{R}\Gamma_c(\mathfrak{p} \cap \mathcal{N}, (\mathbb{R}p_{\mathcal{N},*} \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}})|_{\mathfrak{p} \cap \mathcal{N}}).$$

Finally, there is a W -action on $(H_c^{2d}(T_K^* \mathcal{B}, \mathbb{Q}))^* = \bigoplus_{\mathcal{O} \in K \backslash \mathcal{B}} \mathbb{Q}[\overline{T_{\mathcal{O}}^* \mathcal{B}}]$.

Theorem 12.1 (Tanisaki). *The map:*

$$\mathrm{CC} : K_0(\mathrm{Mod}_K^{\mathrm{Coh}}(\mathcal{D}_{\mathcal{B}})) \rightarrow (H_c^{2d}(T_K^* \mathcal{B}, \mathbb{Q}))^* = H_{2d}^{\mathrm{BM}}(T_K^* \mathcal{B}, \mathbb{Q}),$$

is surjective and W -equivariant.

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